

EVERY n POINTS IN THE PLANE CONTAIN $(m_1 + o(1))n$ POINTS WITH NO UNIT DISTANCE

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ABSTRACT. Let $f(n)$ be the largest integer such that every set of n points in the plane contains $f(n)$ points no two at distance 1, and let m_1 be the supremum of the upper densities of Lebesgue measurable planar sets with no two points at distance 1. Larman and Rogers observed that $f(n) \geq m_1 n$. We prove that $f(n)/n \rightarrow m_1$; that is, $f(n) = (m_1 + o(1))n$. This is the first half of a conjecture of Dúcz and Varga, and with the bound $m_1 \leq 0.247$ of Ambrus, Csiszárík, Matolcsi, Varga and Zsámboki it gives $f(n) \leq (0.247 + o(1))n$. The proof transfers an arbitrary, possibly non-measurable, independent set to a measurable one. Its only non-classical input is the Haar rigidity theorem for wild character laws from the OpenAI preprint *The Euclidean plane is not five-colorable*, which has a machine-checked Lean proof. The whole argument has been formalized in Lean 4 on top of it.

1. INTRODUCTION

For a finite set $P \subset \mathbb{R}^2$ let G_P be its unit-distance graph (vertices P , edges the pairs at Euclidean distance exactly 1) and $\alpha(P)$ its independence number. Erdős [7] asked to estimate

$$f(n) = \min\{\alpha(P) : P \subset \mathbb{R}^2, |P| = n\},$$

and in particular whether $f(n) \geq n/4$ (Problem #1070 in Bloom's database [4]). The Moser spindle gives $f(n) \leq \frac{2}{7}n + O(1)$. Matolcsi, Ruzsa, Varga and Zsámboki [9] proved $f(n) \leq (\frac{1}{4} + o(1))n$ and conjectured that this is sharp. Dúcz and Varga [6] disproved that conjecture by exhibiting a finite unit-distance graph with independence ratio below $1/4$, answering the second question of Erdős negatively; their result has been formalized in Lean by Benkő [3].

For the lower bound, let m_1 be the supremum of the upper densities of measurable sets $A \subset \mathbb{R}^2$ containing no two points at distance 1. Larman and Rogers [8] observed $f(n) \geq m_1 n$; Croft [5] proved $m_1 \geq 0.22936$, and Ambrus, Csiszárík, Matolcsi, Varga and Zsámboki [1] proved $m_1 \leq 0.247$, settling Erdős's question whether $m_1 < 1/4$ (Problem #232 of [4]). Dúcz and Varga [6, Conjecture 1] conjecture that $f(n)/n = m_1 + o(1)$ and that $m_1 = \delta_{\text{Croft}}$. We prove the first half.

Theorem 1.1. $\lim_{n \rightarrow \infty} f(n)/n = m_1$. Moreover $f(n) \geq m_1 n$ for every $n \geq 1$.

Corollary 1.2. $0.22936 n \leq f(n) \leq (0.247 + o(1))n$. In particular $f(n) < n/4$ for all large n .

Theorem 1.1 says that the two natural constants of the problem, the worst finite independence ratio and the best measurable density, coincide; [9] had conjectured that they differ ($1/4$ versus δ_{Croft}). Determining the numerical value of m_1 remains open.

Write

$$\alpha^* = \inf \left\{ \frac{\alpha(P)}{|P|} : P \subset \mathbb{R}^2 \text{ finite, nonempty} \right\}.$$

Theorem 1.1 follows from three statements: $f(n)/n \rightarrow \alpha^*$ with $f(n) \geq \alpha^*n$ (Section 3); $\alpha^* \geq m_1$ (Larman–Rogers, Section 4); and $m_1 \geq \alpha^*$ (Section 5). The last is the new part. It is a single-label version of the transfer in Section 4 of the OpenAI preprint [10], which turns an arbitrary proper colouring into a measurable weak colouring; the one change needed is in how the averaging is done (Remark 5.4).

Formalization. All of the above is formalized in Lean 4 with Mathlib, building on OpenAI’s formalization [11] of the five-colour paper; see Section 6.

2. DENSITIES

For a measurable $A \subset \mathbb{R}^2$ and a family of regions $(W_r)_r$ put

$$\bar{\delta}_W(A) = \limsup_{r \rightarrow \infty} \frac{|A \cap W_r|}{|W_r|},$$

where $|\cdot|$ is Lebesgue measure. We consider four families: discs $B(0, r)$ or squares $[-r, r]^2$, with r ranging over $\mathbb{R}_{>0}$ or over $\mathbb{N}$. Let m_1^W be the supremum of $\bar{\delta}_W(A)$ over measurable unit-distance-free A . The database [4] uses discs with real r .

Remark 2.1. All four values m_1^W equal α^* , so m_1 is unambiguous. Indeed, Proposition 4.1 below gives $m_1^W \leq \alpha^*$ for each family, and Section 5 constructs, for discs and for squares, a measurable unit-distance-free A with $\limsup_{r \in \mathbb{N}} |A \cap W_r|/|W_r| \geq \alpha^*$; a limsup over $r \in \mathbb{R}$ is at least the limsup over $r \in \mathbb{N}$.

3. CONVERGENCE OF $f(n)/n$

Proposition 3.1. $f(n) \geq \alpha^*n$ for every n , and $f(n)/n \rightarrow \alpha^*$.

Proof. If $|P| = n$ then $\alpha(P) \geq \alpha^*n$ by definition; hence $f(n) \geq \alpha^*n$. Fix a finite P_0 with $|P_0| = s$ and $\alpha(P_0) = a$. For $N \geq s$ write $N = ks + r$ with $0 \leq r < s$. Take k translates of P_0 and r single points, placed so that distinct pieces are at distance more than $\text{diam } P_0 + 2$. The union has N points, its unit-distance graph is the disjoint union of the pieces’ graphs, and its independence number is $ka + r$. Hence $f(N) \leq ka + r \leq Na/s + s$, so $\limsup_N f(N)/N \leq a/s$. Taking the infimum over P_0 gives $\limsup_N f(N)/N \leq \alpha^*$. $\square$

4. THE BOUND $\alpha^* \geq m_1$

Proposition 4.1 (Larman–Rogers [8]). *Let A be measurable and unit-distance-free, and let W be any of the four families. Then $\alpha(P) \geq \bar{\delta}_W(A) |P|$ for every finite P . Hence $m_1^W \leq \alpha^*$.*

Proof. Let $P = \{p_1, \dots, p_n\}$, $\rho = \max_j |p_j|$, $\delta = \bar{\delta}_W(A)$. For each x , $(P+x) \cap A$ is an independent set of $G_{P+x} \cong G_P$, so $|(P+x) \cap A| \leq \alpha(P)$. By Tonelli,

$$\alpha(P) |W_r| \geq \int_{W_r} |(P+x) \cap A| dx = \sum_j |A \cap (W_r + p_j)| \geq n(|A \cap W_r| - |W_r \setminus (W_r + p_j)|).$$

For discs and squares, $|W_r \setminus (W_r + p)| \leq 2\pi r |p|$ and $|W_r| \geq \pi r^2$, so $\alpha(P) \geq n|A \cap W_r|/|W_r| - 2n\rho/r$. Letting $r \rightarrow \infty$ along a sequence realizing the limsup gives $\alpha(P) \geq n\delta$. $\square$

5. THE BOUND $m_1 \geq \alpha^*$

5.1. Setting. Identify $\mathbb{R}^2$ with $\mathbb{C}$. Let $E = \overline{\mathbb{Q}} \subset \mathbb{C}$ be the field of algebraic numbers, viewed as a countable discrete additive group, and $K = \{u \in E : |u| = 1\}$, a countable discrete multiplicative group. For $x, y \in E$ we have $|x - y| = 1$ iff $x - y \in K$. K is dense in the unit circle S^1 , since $\frac{(1-t^2)+2it}{1+t^2} \in K$ for $t \in \mathbb{Q}$.

Let $D = \widehat{E}$ be the compact (metrizable, as E is countable) dual group of E ; for $a, z \in E$ and $d \in D$ put $(ad)(z) = d(az)$ and $\psi_z(d) = d(z)$. The continuous characters $C = \{j(\xi) : \xi \in \mathbb{R}^2\}$, $j(\xi)(z) = e^{i\xi \cdot z}$, form a Borel subgroup of D .

Theorem 5.1 (Haar rigidity [10, Thm. 2.3]). *Every K -invariant Borel probability measure ν on D with $\nu(C) = 0$ is Haar measure. Equivalently (by uniqueness of Fourier coefficients on D), $\int_D \psi_z d\nu = 0$ for all $z \in E \setminus \{0\}$.*

The second form is proved in Lean as `OAI.PlaneFiveColor.Spectral.wild_fourier` [11]. We also use the following lemma, whose proof in [10] is a standard spectral argument (spectral theorem for the discrete group E , Bochner's theorem on $\mathbb{R}^2$, uniqueness of Fourier coefficients on D).

Lemma 5.2 ([10, Lemma 4.1]). *Let E act by measure-preserving maps on a standard probability space (Ω, μ) , with Koopman operators T_z on $L^2(\mu)$. Let $\mathcal{H}_c = \{v \in L^2(\mu) : \|T_z v - v\|_2 \rightarrow 0 \text{ as } z \rightarrow 0, z \in E\}$, the limit in the topology of $\mathbb{C}$. Then:*

- (i) $\mathcal{H}_c = L^2(\Omega, \mathcal{F}_c, \mu)$ for a sub- σ -algebra $\mathcal{F}_c$, so the orthogonal projection onto $\mathcal{H}_c$ is $\mathbb{E}(\cdot | \mathcal{F}_c)$;
- (ii) $\mathcal{H}_c$ is the range of the spectral projection of $\{T_z\}$ on C ;
- (iii) the restrictions $T_z|_{\mathcal{H}_c}$ extend to a strongly continuous unitary representation $(T_x)_{x \in \mathbb{R}^2}$ on $\mathcal{H}_c$ preserving the cone of nonnegative functions and fixing $\mathbf{1}$;
- (iv) if unitaries V_u ($u \in K$) satisfy $V_u T_z V_u^{-1} = T_{uz}$, then $V_u \mathcal{H}_c = \mathcal{H}_c$ and V_u commutes with the projection onto $\mathcal{H}_c$.

5.2. An invariant random independent set. Let $\Omega \subset \{0, 1\}^E$ be the set of independent sets of the unit-distance graph on E . Each edge constraint is clopen, so Ω is a closed subset of the compact metrizable space $\{0, 1\}^E$. Translations $s_z \omega = \omega - z$ ($z \in E$) and rotations $\rho_u \omega = u\omega$ ($u \in K$) are homeomorphisms of Ω , and $s_w \rho_u = \rho_u s_{u^{-1}w}$.

We use Følner sequences in E and K : finite sets F_N with $|F_N \Delta (F_N + w)|/|F_N| \rightarrow 0$ for every w (and the multiplicative analogue in K). They exist because E and K are countable abelian groups, hence amenable [2, Thm. G.5.1]. For E one can take explicitly: enumerate $E = \{e_0, e_1, \dots\}$, let $\Lambda_N \cong \mathbb{Z}^{d_N}$ be the subgroup generated by $e_0, \dots, e_N$ (E is torsion-free), and let F_N be a sufficiently large coordinate box in Λ_N .

Lemma 5.3. *There is a Borel probability measure μ on Ω , invariant under all s_z and ρ_u , with $\mu(\{\omega : 0 \in \omega\}) \geq \alpha^*$.*

Proof. The space $\mathcal{P}(\Omega)$ of Borel probability measures is weak-* compact and metrizable (Riesz–Markov and Banach–Alaoglu), and for a homeomorphism g of Ω the map $\nu \mapsto g_* \nu$ is weak-* continuous. The set $\{0 \in \omega\}$ is clopen, so $\nu \mapsto \nu(0 \in \omega)$ is weak-* continuous.

Translations. Each F_N is a finite subset of $\mathbb{R}^2$; choose a maximum independent set $J_N \subset F_N$, so $|J_N| \geq \alpha^* |F_N|$ and $J_N - z \in \Omega$ for all z . Let $\nu_N = |F_N|^{-1} \sum_{z \in F_N} \delta_{J_N - z}$. Then $\nu_N(0 \in \omega) = |J_N|/|F_N| \geq \alpha^*$ and $\|(s_w)_* \nu_N - \nu_N\|_{TV} \leq |F_N \Delta (F_N + w)|/|F_N| \rightarrow 0$. Any weak-* limit point μ_0 is therefore s_w -invariant for every w and satisfies $\mu_0(0 \in \omega) \geq \alpha^*$.

Rotations. Let G_M be Følner sets in K and $\mu_M = |G_M|^{-1} \sum_{u \in G_M} (\rho_u)_* \mu_0$. Each $(\rho_u)_* \mu_0$ is translation invariant, because $(s_w)_* (\rho_u)_* \mu_0 = (\rho_u)_* (s_{u^{-1}w})_* \mu_0 = (\rho_u)_* \mu_0$; and $(\rho_u)_* \mu_0(0 \in \omega) =$

$\mu_0(0 \in \omega)$ since $0 \in u\omega \iff 0 \in \omega$. A weak-* limit point μ of (μ_M) is translation invariant, rotation invariant by the Følner property, and has $\mu(0 \in \omega) \geq \alpha^*$. $\square$

Remark 5.4. In [10, Prop. 4.2] a single colouring is averaged over Følner sets of $E \rtimes K$. Applied to an independent set J , this produces $\mu(0 \in \omega) = \sum_w c(w) \mathbf{1}_J(w) / \sum_w c(w)$ with multiplicities $c(w) = \#\{g \in \Phi : g^{-1}0 = w\}$ that are not constant in w , i.e. a weighted independence ratio, which can be smaller than α^* . Averaging first over translations (where each position is counted once) and then over rotations (which fix 0) avoids this.

5.3. Projection onto the continuous factor. Fix μ from Lemma 5.3. Since Ω is compact metrizable, (Ω, μ) is a standard probability space. Define unitaries on $L^2(\mu)$ by $T_z F(\omega) = F(\omega - z)$ and $V_u F(\omega) = F(u^{-1}\omega)$; then $T_z T_w = T_{z+w}$ and $V_u T_z V_u^{-1} = T_{uz}$. Let $f = \mathbf{1}[0 \in \omega]$. Then $T_z f = \mathbf{1}[z \in \omega]$, $V_u f = f$, $\mathbb{E}f \geq \alpha^*$, and since 0 and $u \in K$ are adjacent, $f \cdot T_u f = 0$ pointwise, so

$$(1) \quad \langle f, T_u f \rangle = 0 \quad (u \in K).$$

Let $p = \mathbb{E}(f \mid \mathcal{F}_c) \in \mathcal{H}_c$ and $b = f - p \in \mathcal{H}_c^\perp$ (Lemma 5.2). Then $0 \leq p \leq 1$ a.e., $\mathbb{E}p = \mathbb{E}f$, and $V_u p = p$, $V_u b = b$. Both p and b are real-valued.

Let λ_b be the spectral measure of b on D : $\langle b, T_z b \rangle = \int_D \psi_z d\lambda_b$. By Lemma 5.2(ii), $\lambda_b(C) = 0$. For $u \in K$,

$$\langle b, T_{uz} b \rangle = \langle b, V_u T_z V_u^{-1} b \rangle = \langle V_u^{-1} b, T_z V_u^{-1} b \rangle = \langle b, T_z b \rangle,$$

so $u_* \lambda_b$ and λ_b have the same Fourier coefficients and λ_b is K -invariant. If $b \neq 0$, Theorem 5.1 applied to $\lambda_b / \|b\|^2$ gives $\langle b, T_z b \rangle = 0$ for $z \neq 0$; this is trivial if $b = 0$. Since T_u preserves $\mathcal{H}_c$ and $\mathcal{H}_c^\perp$, $\langle p, T_u p \rangle = \langle b, T_u b \rangle = 0$. Expanding (1),

$$\langle p, T_u p \rangle = 0 \quad (u \in K).$$

By Lemma 5.2(iii), $(T_x)_{x \in \mathbb{R}^2}$ is a strongly continuous unitary representation on $\mathcal{H}_c \ni p$, so $u \mapsto \langle p, T_u p \rangle$ is continuous and, K being dense in S^1 , vanishes on S^1 . Hence for all $x \in \mathbb{R}^2$ and $u \in S^1$,

$$(2) \quad \int (T_x p)(T_{x+u} p) d\mu = \langle T_x p, T_{x+u} p \rangle = \langle p, T_u p \rangle = 0.$$

Moreover $0 \leq T_x p \leq 1$ a.e. (apply positivity to p and $\mathbf{1} - p$) and $\mathbb{E}[T_x p] = \langle T_x p, \mathbf{1} \rangle = \langle p, T_{-x} \mathbf{1} \rangle = \mathbb{E}p \geq \alpha^*$.

5.4. A measurable field and one sample.

Lemma 5.5. *Let $x \mapsto v_x$ be a continuous map from $\mathbb{R}^2$ to $L^2(\mu)$. There is a jointly measurable $H : \mathbb{R}^2 \times \Omega \rightarrow \mathbb{R}$ with $H(x, \cdot) = v_x$ μ -a.e. for every $x \in \mathbb{R}^2$.*

Proof. Fix a dense sequence (x_k) in $\mathbb{R}^2$ and a representative of each v_{x_k} . For $j \geq 1$ let $k_j(x)$ be the least k with $\|v_{x_k} - v_x\|_2 < 2^{-j}$; it exists by continuity and density, and $x \mapsto k_j(x)$ is Borel because each $x \mapsto \|v_{x_k} - v_x\|_2$ is continuous. Put $H_j(x, \omega) = v_{x_{k_j(x)}}(\omega)$, jointly measurable. For fixed x , $\sum_j \|H_j(x, \cdot) - v_x\|_2 < \infty$, so $H_j(x, \cdot) \rightarrow v_x$ a.e. Let $H = \limsup_j H_j$ where finite and 0 elsewhere. $\square$

Apply Lemma 5.5 to $v_x = T_x p$ and replace H by $\min(\max(H, 0), 1)$. Fix the family W (discs or squares) and integer radii $r \in \mathbb{N}$. By (2) and Tonelli, for every $r \in \mathbb{N}$,

$$\int_\Omega \int_{B(0,r)} \int_{S^1} H(x, \omega) H(x+u, \omega) d\sigma(u) dx d\mu(\omega) = 0,$$

and the inner double integral is nonnegative; so there is a μ -conull Ω_1 on which it vanishes for every $r \in \mathbb{N}$. Let $X_r(\omega) = |W_r|^{-1} \int_{W_r} H(x, \omega) dx \in [0, 1]$. By Tonelli, $\mathbb{E}X_r = \mathbb{E}p \geq \alpha^*$, and by

the reverse Fatou lemma ($X_r \leq 1$), $\mathbb{E}[\limsup_r X_r] \geq \alpha^*$; hence $\mu(\limsup_r X_r \geq \alpha^*) > 0$. Fix $\omega \in \Omega_1$ with $\limsup_r X_r(\omega) \geq \alpha^*$ and set $h = H(\cdot, \omega) : \mathbb{R}^2 \rightarrow [0, 1]$:

$$(3) \quad \int_{B(0,r)} \int_{S^1} h(x)h(x+u) d\sigma dx = 0 \quad (r \in \mathbb{N}), \quad \limsup_{r \in \mathbb{N}} \frac{1}{|W_r|} \int_{W_r} h \geq \alpha^*.$$

5.5. Density points. Let $A_0 = \{h > 0\}$. By (3), $\int_{B(0,r)} \int_{S^1} \mathbf{1}_{A_0}(x)\mathbf{1}_{A_0}(x+u) d\sigma dx = 0$ for all r , and since $h \leq \mathbf{1}_{A_0}$, $\limsup_{r \in \mathbb{N}} |A_0 \cap W_r|/|W_r| \geq \alpha^*$. Let A be the set of Lebesgue density points of A_0 . It is Borel, and $|A \triangle A_0| = 0$ by the Lebesgue density theorem, so A has the same densities as A_0 .

Lemma 5.6. *A contains no two points at distance 1.*

Proof. Suppose $x, y = x + u \in A$ with $|u| = 1$. For $0 < \eta < 1$ let x' be uniform on $B(x, \eta)$ and, independently, u' uniform on the arc $\{e^{i\theta}u : |\theta| < \eta/2\}$, so $|u' - u| < \eta$. Then $\mathbb{P}(x' \notin A_0) = |B(x, \eta) \setminus A_0|/|B(x, \eta)| \rightarrow 0$. Given u' , $x' + u'$ is uniform on $B(x + u', \eta) \subset B(y, 2\eta)$, so $\mathbb{P}(x' + u' \notin A_0) \leq |B(y, 2\eta) \setminus A_0|/(\pi\eta^2) = 4|B(y, 2\eta) \setminus A_0|/|B(y, 2\eta)| \rightarrow 0$. So for small η , $\mathbb{P}(x' \in A_0, x' + u' \in A_0) > 0$. The law of (x', u') has a bounded density with respect to $dx \otimes \sigma$ and is supported in $B(0, r) \times S^1$ for an integer $r > |x| + 1$, contradicting the vanishing of $\int_{B(0,r)} \int_{S^1} \mathbf{1}_{A_0}(x)\mathbf{1}_{A_0}(x+u) d\sigma dx$. $\square$

Thus A is a measurable unit-distance-free set with $\limsup_{r \in \mathbb{N}} |A \cap W_r|/|W_r| \geq \alpha^*$, for W = discs and for W = squares. This proves $m_1^W \geq \alpha^*$ for all four conventions, completing Remark 2.1 and, with Propositions 3.1 and 4.1, Theorem 1.1. Corollary 1.2 follows from [5, 1].

6. FORMALIZATION

The repository [12] contains a Lean 4 project (toolchain v4.34.1, Mathlib d13f23b). The directory `OAI/` is a verbatim copy of the 70-file import closure of OpenAI's five-colour theorem from [11] (Apache-2.0). Our files define

- `indepNum P` (independence number of the unit-distance graph on a `Finset C`),
- `f n`, the minimum of `indepNum P` over n -element P ,
- `upperDensity` (squares $[-m, m]^2$, $m \in \mathbb{N}$) and `m1`,

and prove `f_div_tendsto_m1` ($f(n)/n \rightarrow m_1$) and `m1_mul_le_f` ($m_1 n \leq f(n)$). `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for both, and for `wild_fourier`. The Lean proof follows Sections 3–5; the Følner sets in E are constructed explicitly as above, and the weak-* limits are realized by ultrafilter limits of positive functionals and the Riesz–Markov theorem.

ACKNOWLEDGEMENTS

The negative answer to the second question of Erdős is due to Dúcz and Varga [6]. Theorem 5.1 and the transfer template are from [10]. AI tools were used in this work: the observation that [10, Section 4] applies to a single colour class, the write-up and the Lean formalization were produced with Claude (Anthropic), and the manuscript was audited with GPT (OpenAI).

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